

Assignment Brief – Individual AI Security Audit and Vulnerability Assessment Information Retrieval and Web Analytics (IT3041)

Lecturer in Charge:

Mr. Samadhi Chathuranga Rathnayake

Assignment Overview

Building an intelligent AI system is only one aspect of modern software engineering. Before deployment, every AI system must undergo rigorous security, privacy, ethical, and reliability evaluations to ensure that it behaves safely under both normal and adversarial conditions. This assignment requires each student to independently perform a professional vulnerability assessment of the **Agentic AI system developed during the group assignment**. Students will act as **AI Security Analysts (Red Team Members)** and evaluate the system from a designated security perspective.

The objective is not to redesign or improve the existing system, but to critically assess its robustness by identifying vulnerabilities, evaluating associated risks, and recommending appropriate mitigation strategies.

Each student will submit an independent **AI Vulnerability Assessment Report** and defend their findings during an individual viva examination.

Learning Objectives

Upon successful completion of this assignment, students should be able to:

- Perform systematic security evaluations of AI systems.
- Apply AI Red Teaming techniques to identify system vulnerabilities.
- Assess AI systems against Responsible AI principles.
- Evaluate the robustness of Large Language Model (LLM)-based applications.
- Conduct structured risk assessments.
- Recommend practical mitigation strategies.
- Produce professional vulnerability assessment reports.

Assignment Structure

The same project groups formed for the Group Assignment will be used.

Each student within a group will be assigned **one primary evaluation specialization** by the lecturer. While each student will focus on a different specialization, every student is expected to understand the complete system architecture and operation.

Each student must independently perform all required testing related to the assigned specialization and prepare an individual report.

Individual Evaluation Specializations

Student 1

Prompt Injection and Jailbreak Analysis

Evaluate the AI system against prompt-based attacks.

Suggested evaluation areas include:

- Prompt Injection
- Jailbreak Attempts
- Prompt Leakage
- Instruction Override
- Prompt Manipulation
- Prompt Robustness

Expected outcome:

A professional vulnerability assessment identifying prompt-related weaknesses and recommended countermeasures.

Student 2

Privacy and Data Leakage Assessment

Evaluate the protection of user data and confidential information.

Suggested evaluation areas include:

- Sensitive Information Leakage
- Personally Identifiable Information (PII) Exposure
- Conversation Memory Leakage
- Authentication Weaknesses
- User Data Protection
- Privacy Compliance

Expected outcome:

A professional privacy and data protection assessment.

Student 3

Responsible AI and Bias Assessment

Evaluate the ethical behaviour of the AI system.

Suggested evaluation areas include:

- Hallucinations
- Bias
- Toxic Responses
- Fairness
- Transparency
- Explainability
- Harmful Content Generation

Expected outcome:

A Responsible AI compliance and vulnerability assessment.

Student 4

Information Retrieval and Security Assessment

Evaluate the security and reliability of the Information Retrieval components.

Suggested evaluation areas include:

- Retrieval Accuracy
- Retrieval Manipulation
- Hallucination due to Retrieval
- Source Reliability
- Authentication
- Authorization
- API Security
- Communication Protocol Security

Expected outcome:

A technical security assessment of the Information Retrieval pipeline.

Testing Requirements

Each student must perform **a minimum of 15 independent test cases** related to the assigned specialization.

Each test case should clearly include:

- Test Objective
- Input / Attack Scenario
- Expected Behaviour
- Actual Behaviour
- Evidence (Screenshots or Logs)
- Observations
- Conclusion

Risk Classification

Each identified vulnerability must be classified using one of the following severity levels:

- Critical
- High
- Medium
- Low
- Informational

Students must provide clear technical justification for every assigned severity level.

Individual Deliverables

Each student must submit independently.

1. Vulnerability Assessment Report (80 Marks)

The report should include the following sections:

Executive Summary

Provide a brief overview of the assessment, objectives, and major findings.

Scope of Testing

Describe:

- System evaluated
- Assigned specialization
- Components evaluated
- Scope limitations

Evaluation Methodology

Describe:

- Testing methodology
- Evaluation process
- Tools used
- Testing environment
- Evaluation criteria

Test Cases Performed

Document all test cases with supporting evidence.

Each test case should include:

- Test ID
- Objective
- Input
- Expected Result
- Actual Result
- Evidence
- Outcome

Vulnerabilities Identified

For each identified vulnerability include:

- Vulnerability Description
- Evidence
- Impact
- Likelihood
- Severity
- Risk Level
- Technical Explanation

Risk Assessment

Summarize all identified vulnerabilities using a risk matrix.

Example:

Vulnerability	Impact	Likelihood	Risk Level
Prompt Injection	High	High	Critical

Mitigation Strategies

Recommend practical solutions for each identified vulnerability.

Reflection

Briefly discuss:

- Challenges encountered
- Lessons learned
- Recommendations for future improvements

2. Individual Viva (20 Marks)

Students will be required to explain:

- The testing methodology used.
- The rationale behind the selected test cases.
- Why attacks succeeded or failed.
- The technical reasoning behind identified vulnerabilities.
- Justification of the assigned risk levels.
- Proposed mitigation strategies.
- Responsible AI implications.

Required Testing Evidence

Students should perform **at least 15 different test cases** within their assigned specialization.

Examples include:

Student 1

- 15 prompt injection or jailbreak attempts.

Student 2

- 15 privacy or data leakage scenarios.

Student 3

- 15 Responsible AI, bias, or hallucination evaluations.

Student 4

- 15 Information Retrieval or security evaluations.

Screenshots, logs, or other supporting evidence must be included in the report.

Assessment Breakdown

Component	Marks
Vulnerability Assessment Report	50
Individual Viva	50
Total	100

Report Evaluation Rubric (50 Marks)

Criteria	Marks
Testing Methodology and Coverage	10
Quality of Vulnerability Analysis	10
Risk Assessment and Severity Classification	10
Mitigation Strategies	10
Report Quality, Evidence, and Technical Writing	10
Total	50